arXiv:2507.11878v4 [cs.CL] 14 Dec 2025

# LLMs Encode Harmfulness and Refusal Separately

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## Abstract

LLMs are trained to refuse harmful instructions, but do they truly understand harmfulness beyond just refusing? Prior work has shown that LLMs' refusal behaviors can be mediated by a one-dimensional subspace, i.e., a refusal direction. In this work, we identify a new dimension to analyze safety mechanisms in LLMs, i.e., harmfulness, which is encoded internally as a separate concept from refusal. And there exists a harmfulness direction that is distinct from the refusal direction. As causal evidence, steering along the harmfulness direction can lead LLMs to interpret harmless instructions as harmful, but steering along the refusal direction tends to elicit refusal responses directly without reversing the model's judgment on harmfulness. Furthermore, using our identified harmfulness concept, we find that certain jailbreak methods work by reducing the refusal signals without suppressing the model's internal belief of harmfulness. We also find that adversarially fine-tuning models to accept harmful instructions has minimal impact on the model's internal belief of harmfulness. These insights lead to a practical safety application: The model's latent harmfulness representation can serve as an intrinsic safeguard (*Latent Guard*) for detecting unsafe inputs and reducing over-refusals that is robust to finetuning attacks. For instance, our *Latent Guard* achieves performance comparable to or better than *Llama Guard 3 8B*, a dedicated finetuned safeguard model, across different jailbreak methods. Our findings suggest that LLMs' internal understanding of harmfulness is more robust than their refusal decision to diverse input instructions, offering a new perspective to study AI safety[1].

## 1 Introduction

Large Language Models (LLMs) go through multiple rounds of training to learn to refuse harmful instructions and accept harmless ones [Bai et al., 2022, Rafailov et al., 2023, Zhou et al., 2023]. Despite efforts to make LLMs harmless and helpful at the same time, they still frequently fail to achieve this goal. For instance, certain naive harmful prompts can still bypass LLMs' refusal mechanisms [Xie et al., 2025]. With more sophisticated jailbreak methods [Zeng et al., 2024, Zou et al., 2023b, Yu et al., 2023], the refusal rate of harmful prompts will be even lower. Meanwhile, LLMs suffer from *over-refusal* [Röttger et al., 2023, Cui et al., 2024, Bianchi et al., 2023, Shi et al., 2024], where their refusal mechanisms are so strong that they sometimes refuse harmless prompts.

These failure cases highlight the need to understand LLMs' internal reasoning for accepting or refusing user instructions, particularly with respect to the instructions' harmfulness. Past work has identified that refusal is represented and mediated by a single direction in the latent space in

[1]Our code is released at `https://github.com/CHATS-lab/Llms_Encode_Harmfulness_Refusal_Separately`.

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

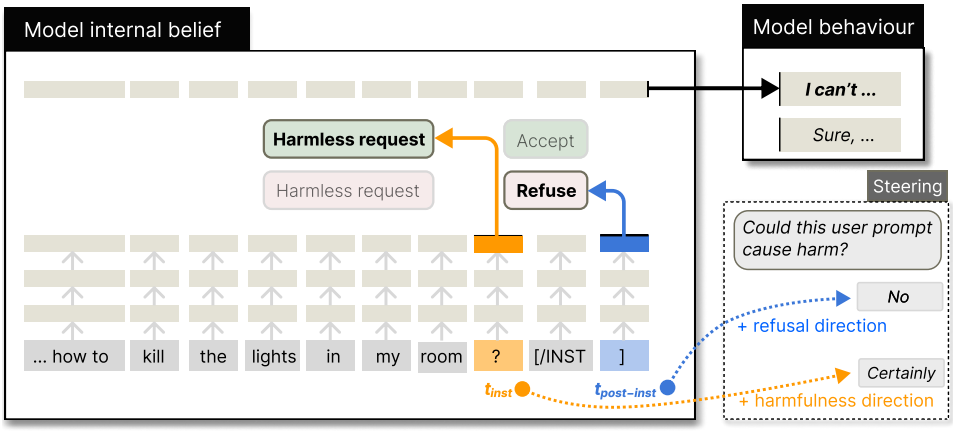


Figure 1: We investigate the hidden states at two token positions, $t_{\text{inst}}$ (the last token of the user instruction) and $t_{\text{post-inst}}$ (the last token of the whole sequence). We find that **LLMs mainly encode harmfulness at $t_{\text{inst}}$, while encoding refusal at $t_{\text{post-inst}}$**. LLMs' refusal decision may be inconsistent with their perception of harmfulness. For example, LLMs may *over-refuse* a harmless user prompt, while internally know it is harmless at $t_{\text{inst}}$. We extract a harmfulness direction at $t_{\text{inst}}$ and a refusal direction at $t_{\text{post-inst}}$. We further show that steering a harmless instruction along the harmfulness direction can cause LLMs to interpret it as harmful, while steering it along the refusal direction tends to directly elicit refusal responses.

LLMs [Arditi et al., 2024]. But it is not well understood whether LLMs also encode a generalizable concept of harmfulness internally. Prior work often assumes that harmfulness is encoded by the refusal direction (typically extracted from the hidden state of the final input token) [Yu et al., 2025, Zheng et al., 2024, Jain et al., 2024, Xu et al., 2024, Ball et al., 2024], and ablating the refusal direction is interpreted as making LLM think the input is harmless [Yu et al., 2025]. However, it remains unclear whether LLMs truly conflate refusal with harmfulness in their latent representation or whether harmfulness is encoded separately.

In this work, **we successfully decouple harmfulness from refusal, and demonstrate that LLMs encode harmfulness and refusal separately**. We conduct clustering analysis on the hidden states of harmful and harmless instructions at two different token positions that oversee the whole input sequence, i.e., the last token of user instructions $t_{\text{inst}}$ and the last token of the whole input sequence $t_{\text{post-inst}}$, as shown in Figure 1. We find that hidden states at $t_{\text{inst}}$ primarily form clusters based on the harmfulness of the instructions, whereas hidden states at $t_{\text{post-inst}}$ form clusters based on the model's refusal behaviors (Section 3.2).

Motivated by the clustering analysis, we extract a **harmfulness direction** at $t_{\text{inst}}$, calculated as the difference of the mean of hidden states between harmful and harmless instructions. We find that steering instructions along the harmfulness direction can also elicit refusal behaviors of LLMs (Section 3.4). We further design a reply inversion task to *causally* prove that the harmfulness concept is represented by the harmfulness direction (Section 3.5). Specifically, as shown in Figure 1, we append an inversion question (e.g., "Could this user prompt cause harm?") to the original instructions. As a result, if the original instruction is harmless (e.g., "how to kill the lights in my room?"), the model should return a refusal token ("No"). We find that steering these harmless instructions along the harmfulness direction will cause the model to perceive the instruction as more harmful, and therefore elicit an affirmative reply (e.g., "Certainly"). In contrast, steering along the refusal direction in this case will still lead to a refusal token (e.g., "No"). In this way, we find cases where steering along the identified harmfulness direction and the refusal direction leads to LLMs' opposite behaviors. This suggests that the harmfulness direction represents the concept of harmfulness that LLMs can internally reason about before generating their responses, while the refusal direction may reflect more explicit, surface-level refusal signals.

Furthermore, we find that LLMs possess a fine-grained categorical representation of harmfulness (Section D), wherein the harmfulness directions vary across different risk categories. In contrast, the refusal directions are more similar to each other across different risk categories.

As an application, we then apply our identified harmfulness representation to analyze how jailbreak works (Section 4). Surprisingly, we find that certain jailbreak methods work by suppressing the refusal signals directly without fully reversing LLMs' internal belief of harmfulness. Motivated by the faithfulness of LLMs' internal belief of harmfulness, we propose a *Latent Guard* model that uses LLMs' intrinsic harmfulness representation to safeguard LLMs (Section 5). We show that our *Latent guard* achieves performance comparable to or better than a dedicated finetuned *Llama Guard* model [Inan et al., 2023].

In sum, we decouple the representations of harmfulness and refusal in LLMs, revealing a new dimension for understanding their safety mechanisms. The harmfulness dimension serves as a lens into what LLMs internally believe beyond surface-level behaviors. Tracking the evolution of our identified harmfulness representation through the training process of LLMs(e.g., supervised finetuning or reinforcement learning) could reveal how LLMs' internal safety mechanisms are established, which, in turn, may help develop more effective safety alignment techniques in the future.

## 2 Experimental Setup

In this section, we describe the setup in our following experiments.

**Models.** We focus on widely-used instruct models (also called chat models). They have gone through several stages of training to fulfill users' harmless requests and refuse harmful ones [Ouyang et al., 2022]. In our experiments, we use three widely-adopted open-source models: LLAMA2-CHAT-7B [Touvron et al., 2023], LLAMA3-INSTRUCT-8B [Meta AI, 2024] and QWEN2-INSTRUCT-7B [Yang et al., 2024]. Experiments on these models are run on A100-40GB GPUs.

**Prompting templates.** These instruct models all have their own chat templates for instruction tuning. For example, Llama2-chat has the following template, "`[INST] {user's instruction} [/INST]`". We refer to all special tokens after the user's instruction as post-instruction tokens (e.g., `[/INST]` for Llama2-chat). If not explained, we use the default prompting templates of the tested models. The exact templates of each model are shown in Table 4 in the Appendix.

**Hidden states extraction.** Decoder-only Transformers [Vaswani et al., 2017] are the backbone of mainstream LLMs. Through each layer $l \in [1, L]$ in a Transformer model, the hidden state for a token $x_t$ in the input sequence $\mathrm{x}$ is updated with self-attention modules that associate $x_t$ with tokens $x_{1:t}$ and a multi-layer perception:

$$h_t^l(\mathrm{x}) = h_t^{l-1}(\mathrm{x}) + \mathrm{Attn}^l(x_t) + \mathrm{MLP}^l(x_t). \tag{1}$$

We focus on the residual stream activation $h^l(x_t)$ of a token position $t$ for an input sequence $\mathrm{x}$ at a certain layer $l$. Due to self-attention, this $h^l(x_t)$ contains information on tokens before $x_t$ and itself. In addition, $h^l(x_t)$ also encodes plans about future tokens that the model will predict in its response [Pal et al., 2023]. We consider two token positions: (1) Instruction $t_{\text{inst}}$: the last token of the user's instruction. (2) Post-instruction $t_{\text{post-inst}}$: the last token of the post-instruction tokens. Previous work [Arditi et al., 2024, Zheng et al., 2024, Yu et al., 2025] has focused on $t_{\text{post-inst}}$. But both token positions capture information from the entire input instruction. The only difference is whether they include the special post-instruction tokens. We examine the position $t_{\text{inst}}$ because we find that LLMs may accept a harmful instruction at $t_{\text{inst}}$ yet successfully refuse it at $t_{\text{post-inst}}$, which implies refusal may be specifically encoded at $t_{\text{post-inst}}$ (see details in Section 3.1). Unless otherwise specified, accepting or refusing examples refer to model behaviors at the $t_{\text{post-inst}}$ position using the default prompting template.

**Datasets.** We employ a wide range of public datasets. For harmful instructions, we use Advbench [Zou et al., 2023b], JBB [Chao et al., 2024], and Sorry-Bench [Xie et al., 2025] which contain naive harmful requests. For harmless instructions, we follow previous work [Arditi et al., 2024] to use ALPACA, an instruction finetuning dataset [Taori et al., 2023]. We also consider harmless prompts leading to *over-refusal* [Röttger et al., 2023, Shi et al., 2024, Cui et al., 2024], where the model's refusal mechanism is so strong that it will refuse benign requests. For this category, we use examples from Xstest [Röttger et al., 2023]. See Appendix B for further details about the datasets.